

## Supplementary table

Supplemental Table 1. A comparison of studies predicting side-effects, DTIs, or using networks in either task.

| Study | Prediction Goal                                                                        | Using molecular interaction network | Comparison to our study                                                                                                                                                                                                                              | Performance metric                                                                                                                                                                                                                                                                                         |
|-------|----------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [1]   | Predict new DTIs                                                                       | No                                  | Infers new DTIs using side-effect similarity vs measuring documented DTIs for their utility in predicting underpredicted side effects                                                                                                                | Specificity and sensitivity for identifying if two drugs shared targets. They provided a plot of sensitivity vs specificity and from the image we can estimate their values at a few thresholds:<br>sens = 1, spec = ~0.1<br>sens = ~0.2, spec = ~0.3<br>sens = ~0.1, spec = ~0.6<br>sens = 0, spec = ~0.8 |
| [2]   | Predicting new DTIs                                                                    | Yes                                 | A review paper about the utility of networks to “recommend” new DTIs based on connections between drugs without using structural or negative sample data                                                                                             | N/A                                                                                                                                                                                                                                                                                                        |
| [3]   | Predicting new DTIs                                                                    | Yes                                 | Leveraged connections among drugs, diseases, and genes (like us) in a PPI network algorithm and predicted new DTIs using a network-ranking. Did not compare the predictive utility of different DTI sources.                                         | ROC = 0.95<br>Mean average precision = 0.23                                                                                                                                                                                                                                                                |
| [4]   | Identifying common pathways associated with side effects                               | Yes                                 | Developed KNIME to connect drugs to targets and to protein interactors and uses enrichment to identify relevant pathways. Uses a similar formalism to PathFX but does not compare DTI sources and is restricted to drug-induced liver injury (DILI). | N/A<br>Provided description of the workflow regarding DILI                                                                                                                                                                                                                                                 |
| [5]   | Identifying and derisking drug targets for development                                 | Yes                                 | A review paper that outlines the opportunities for network-based analyses to enhance quantitative systems pharmacology efforts to identify good targets for development                                                                              | N/A                                                                                                                                                                                                                                                                                                        |
| [6]   | Predicting drug side effects from target-gene variation in side-effect-affected organs | No                                  | They demonstrated that for organ systems where drug target genes are genetically linked to phenotypes affecting that organ, there is an increased risk of a drug causing a side effect in that organ.                                                | Odd-ratio = 1.8                                                                                                                                                                                                                                                                                            |
| [7]   | Predicting DTIs from neural networks (NN)                                              | No                                  | Introduced the DeepPurpose library for predicting DTIs from SMILES strings. Did not evaluate utility of DTI sources.                                                                                                                                 | MSE = 0.242 – 0.282 across different NN architectures                                                                                                                                                                                                                                                      |
| [8]   | Predicting DTIs to expedite drug repurposing                                           | No                                  | Represents drugs as graphs and use graph neural networks to predict drug-target affinity. Does not compare predictive utility of DTI sources.                                                                                                        | MSE = 0.880 – 0.893 depending on graph NN variants.                                                                                                                                                                                                                                                        |

## References:

1. Campillos M, Kuhn M, Gavin A-C, Jensen LJ, Bork P. Drug Target Identification Using Side-Effect Similarity. *Science*. 2008;321: 263–266. doi:10.1126/science.1158140
2. Wu Z, Li W, Liu G, Tang Y. Network-Based Methods for Prediction of Drug-Target Interactions. *Frontiers in Pharmacology*. 2018;9: 1134. doi:10.3389/fphar.2018.01134
3. Zhou M, Chen Y, Xu R. A Drug-Side Effect Context-Sensitive Network approach for drug target prediction. *Bioinformatics*. 2018;35: 2100–2107. doi:10.1093/bioinformatics/bty906
4. Füzi B, Malik-Sheriff RS, Manners EJ, Hermjakob H, Ecker GF. KNIME workflow for retrieving causal drug and protein interactions, building networks, and performing topological enrichment analysis demonstrated by a DILI case study. *Journal of Cheminformatics*. 2022;14: 37. doi:10.1186/s13321-022-00615-6
5. Chelliah V, Graaf PH van der. Model-informed target identification and validation through combining quantitative systems pharmacology with network-based analysis. *CPT: Pharmacometrics & Systems Pharmacology*. 2022;11: 399–402. doi:10.1002/psp4.12766
6. Nguyen PA, Born DA, Deaton AM, Nioi P, Ward LD. Phenotypes associated with genes encoding drug targets are predictive of clinical trial side effects. *Nature Communications*. 2019;10: 1579. doi:10.1038/s41467-019-09407-3
7. Huang K, Fu T, Glass LM, Zitnik M, Xiao C, Sun J. DeepPurpose: A Deep Learning Library for Drug-Target Interaction Prediction. *Bioinformatics*. 2020;36: btaa1005-. doi:10.1093/bioinformatics/btaa1005
8. Nguyen T, Le H, Quinn TP, Nguyen T, Le TD, Venkatesh S. GraphDTA: predicting drug–target binding affinity with graph neural networks. *Bioinformatics*. 2020;37: 1140–1147. doi:10.1093/bioinformatics/btaa921